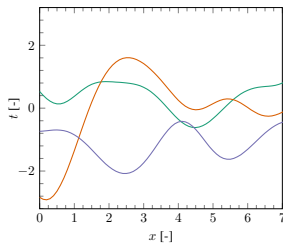
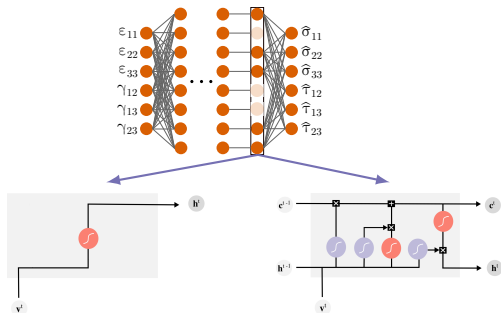


Machine learning regression

Discarding the micromodel and mapping strains to stresses directly:

- Deterministic and probabilistic, parametric and non-parametric models

$$p(t|x) = \mathcal{N}(y(x, \mathbf{w}), \beta) \Rightarrow \sigma = \hat{\sigma}_{\mathbf{w}}(\varepsilon, \alpha) + \text{noise}$$



$$p(\mathbf{t}) = \mathcal{N}(\mathbf{t}|\mathbf{0}, \mathbf{K})$$

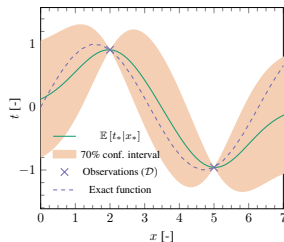
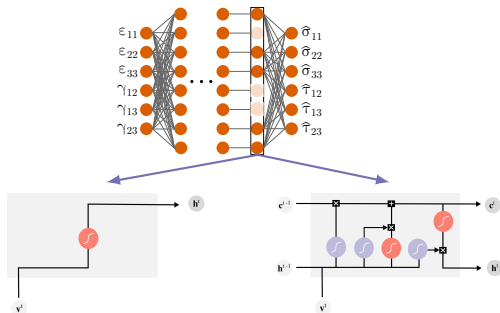
$$K_{pq} = k(\mathbf{x}_p, \mathbf{x}_q) = \sigma_f^2 \exp\left(-\frac{1}{2\ell^2} \|\mathbf{x}_p - \mathbf{x}_q\|^2\right) + \sigma_n^2$$

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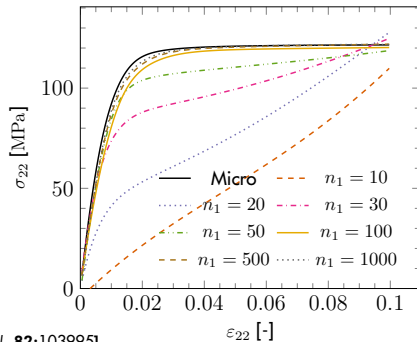
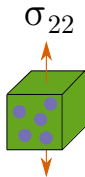
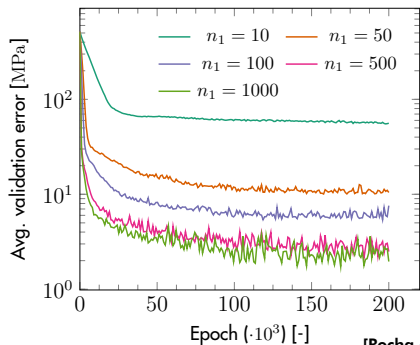
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Feedforward neural nets - Uniaxial stress

Universal approximator, but no upper bound for variance:

- Speed-ups of about **70000 times** for FE² simulations
- No guarantee that even the straightforward elastic behavior is captured

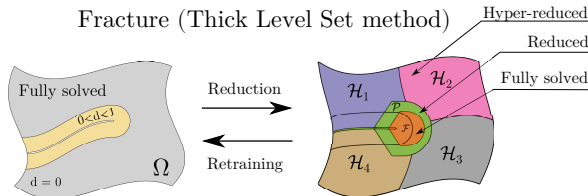
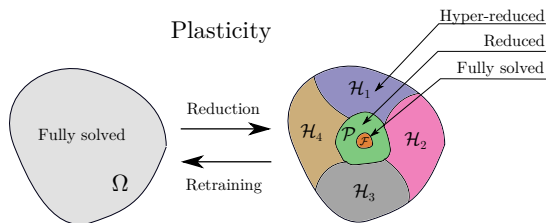


[Rocha et al. (2020), *Eur J Mech A/Sol*, **82**:103995]

Active learning with hyper-reduction

Adaptive hyper-reduction scheme without offline training:

- First step fully solved $\Rightarrow$ POD reduction $\Rightarrow$ Hyper-reduction
- Adaptivity:** Deviation from equilibrium in original space $\Rightarrow$ retraining



Active learning with GPs

Adaptive construction of the constitutive database:

- The model naturally concentrates sampling effort where it is needed

